Notes on Week2

Aodi

1 Effects of Causes

Definition: This type of causal analysis focuses on measuring the outcomes or changes triggered by a specific cause (intervention or exposure). In other words, it examines what kind of "effects" a particular "cause" brings about.

Core Question: If a specific cause occurs (e.g., a policy change, use of a drug), what impact will it have on the outcome?

Examples:

- *Policy Intervention:* Investigating whether increasing the minimum wage reduces poverty.
- *Medical Experiment:* Studying whether a specific drug lowers blood pressure.

Typical Methods: Experimental designs (e.g., randomized controlled trials) or observational studies (e.g., regression analysis) to estimate causal effects.

2 Causes of Effects

Definition: This type of causal analysis seeks to identify the root causes of a particular outcome. The emphasis is on tracing back and understanding "why a result happened."

Core Question: What factors or conditions caused a specific outcome?

Examples:

- *Social Phenomenon:* Why are some countries more prone to populist politics?
- *Personal Behavior:* Analyzing whether a student's grade decline is due to insufficient study time or family pressure.

Typical Methods: Case studies or inductive reasoning to identify potential causes.

3 Deterministic and Probabilistic Causation

3.1 Deterministic Causation

Definition: In deterministic causation, a specific cause inevitably leads to a specific outcome, i.e., "if the cause exists, the effect will always occur."

Example: If water is heated to 100°C (cause), it will boil (effect).

Limitations: Such causation is rare in social sciences, where most phenomena are not driven by single factors.

3.2 Probabilistic Causation

Definition: In probabilistic causation, a specific cause increases the likelihood of an outcome but does not guarantee it.

Example: Smoking (cause) increases the probability of lung cancer (effect), but not all smokers develop lung cancer.

Significance: Probabilistic causation is prevalent in social sciences and medical research.

4 Necessary and Sufficient Conditions

4.1 Necessary Condition

Definition: A necessary condition is a prerequisite for an outcome to occur. If the condition is absent, the outcome cannot happen.

Example: To vote in British Columbia (BC), one must be a BC resident. Without BC residency, voting is impossible.

Key Point: Necessary conditions act as a "threshold." If unmet, the outcome will not occur.

4.2 Sufficient Condition

Definition: A sufficient condition guarantees that the outcome will occur. If this condition is met, the outcome is certain.

Example: To vote in BC, a person must meet the following conditions:

- Be a Canadian citizen,
- Be at least 18 years old, and
- Have resided in BC for at least 6 months before the election.

If these conditions are met, voting eligibility is ensured.

Key Point: Sufficient conditions act as a "guarantee." If satisfied, the outcome will occur.

5 Key Questions for Understanding

5.1 Effects of Causes vs. Causes of Effects

- Are you trying to determine what outcomes a specific cause produces, or are you exploring the root causes of a particular outcome?

5.2 Necessary and Sufficient Conditions

- Necessary conditions are like a lock that requires the correct key (necessary condition) to open.
- Sufficient conditions are like a guarantee—a pre-opened lock that ensures access.